COLLAGE OF CLIMAX

GROUP 4 ASSIGNMENT

TITLE: Introduction to cell based therapy

Member Name Sign

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1. Introduction to Cell-Based Therapy

Cell-based therapy is a medical treatment in which living cells are introduced into a patient's body to repair, replace, or regenerate damaged tissues and organs, or to treat diseases. Unlike traditional medicines that use chemicals or proteins, cell-based therapies use living cells as the therapeutic agent.

Objectives of Cell-Based Therapy

- Repair damaged tissues.
- Replace diseased or dead cells.
- Restore normal body function.
- Treat chronic and genetic diseases.
- Improve healing after injury.

Types of Cell-Based Therapy

- ✓ Stem cell therapy
- ✓ Immune cell therapy (such as CAR-T cell therapy)
- ✓ Tissue engineering
- ✓ Cell transplantation

Applications

- Diabetes
- Parkinson's disease
- Heart disease
- Spinal cord injury
- Blood disorders
- Cancer treatment

2. Stem Cell Therapy

Stem cell therapy is a type of cell-based therapy that uses stem cells to repair, regenerate, or replace damaged cells and tissues.

Stem cells are unique because they can:

- Self-renew (make more stem cells)
- Differentiate into specialized cells like muscle cells, nerve cells, or blood cells

Types of Stem Cells

A. Embryonic Stem Cells (ESCs)

- ✓ Obtained from early embryos.
- ✓ Can develop into almost any cell type.
- ✓ Highly versatile but have ethical concerns.

B. Adult Stem Cells

- Found in bone marrow, fat tissue, blood, and other organs.
- Usually form only specific types of cells.
- Widely used in medicine.

C. Induced Pluripotent Stem Cells (iPSCs)

- Adult cells genetically reprogrammed into stem-cell-like cells.
- Reduce ethical concerns.

Uses of Stem Cell Therapy

- Bone marrow transplantation
- Leukemia treatment
- Tissue regeneration
- Heart repair after heart attack
- Parkinson's disease
- Diabetes research
- Burn treatment

Advantages

- ✓ Can regenerate damaged tissues.
- ✓ May reduce the need for organ transplantation.
- ✓ Can treat previously incurable diseases.
- ✓ Personalized treatment is possible.

Limitations(Disadvantages)

- ✓ High cost.
- ✓ Risk of immune rejection.
- ✓ Ethical concerns.
- ✓ Risk of tumor formation.
- ✓ Requires specialized laboratories.

3.Transplant Rejection

Transplant rejection is the immune system's response against transplanted cells, tissues, or organs because it recognizes them as foreign.

The immune system attacks the transplanted tissue, causing damage or failure.

Types of Transplant Rejection

A. Hyperacute Rejection

- ➢ Occurs within minutes to hours.
- ➢ Caused by pre-existing antibodies.
- ➢ Very severe.

B. Acute Rejection

- ✓ Occurs within days to months.
- ✓ Caused by activation of T cells.
- ✓ Often treatable with immunosuppressive drugs.

C. Chronic Rejection

- Occurs over months or years.
- Causes gradual loss of transplant function.
- Difficult to reverse.

Causes

- Genetic differences between donor and recipient.
- Blood group incompatibility.
- Human Leukocyte Antigen (HLA) mismatch.
- Immune system activation.

Prevention

- Tissue matching.
- Blood group matching.
- Immunosuppressive drugs.
- Regular medical monitoring.

Common Immunosuppressive Drugs Cyclosporine

- ✓ Tacrolimus
- ✓ Prednisone
- ✓ Azathioprine
- ✓ Mycophenolate mofetil

4. Cell Encapsulation Technology

Cell encapsulation technology is a biomedical technique in which living cells are enclosed within a semipermeable protective membrane before being implanted into the body.

The membrane:

- Protects cells from immune attack.
- Allows nutrients and oxygen to enter.

- Allows therapeutic molecules to leave.
- Prevents immune cells from reaching the encapsulated cells.

Types

Microencapsulation

- ✓ Tiny capsules containing one or several cells.
- ✓ Commonly uses alginate.

Macroencapsulation

- Larger devices containing many cells.
- Easier to remove if necessary.

Materials Used

- Alginate
- Chitosan
- Collagen
- Polyethylene glycol (PEG)
- Hydrogel polymers

Applications

- ✓ Diabetes treatment (insulin-producing cells)
- ✓ Parkinson's disease
- ✓ Liver cell transplantation
- ✓ Cancer therapy
- ✓ Gene therapy

Advantages

- Protects transplanted cells.
- Reduces transplant rejection.
- May reduce the need for lifelong immunosuppressive drugs.

- Allows long-term cell survival.

Limitations

- Limited oxygen supply to cells.
- Capsule degradation over time.
- Difficult large-scale production.
- Possible fibrosis around capsules.

5. Production of Recombinant DNA (rDNA) Drugs

Recombinant DNA (rDNA) drugs are medicines produced by inserting a desired gene into a host organism (such as bacteria, yeast, or mammalian cells), enabling it to manufacture therapeutic proteins.

This process uses recombinant DNA technology.

Steps in Production

1. Identify the therapeutic gene.
2. Isolate the DNA.
3. Insert the gene into a plasmid (vector).
4. Transfer the vector into host cells.
5. Grow host cells in bioreactors.
6. Produce the therapeutic protein.
7. Purify the protein.
8. Test quality and safety.

Common Host Cells

- *Escherichia coli* (bacteria)
- Yeast
- Chinese Hamster Ovary (CHO) cells
- Mammalian cell cultures

Examples of rDNA Drugs

- Human insulin
- Human growth hormone
- Erythropoietin (EPO)
- Interferons
- Monoclonal antibodies
- Clotting factors (Factor VIII)
- Tissue plasminogen activator (tPA)

6. Formulation of Recombinant DNA Drugs

Formulation is the process of combining the purified recombinant drug with suitable inactive ingredients (excipients) to produce a stable, safe, and effective dosage form.

Goals

- Improve stability.
- Prevent degradation.
- Maintain biological activity.
- Extend shelf life.
- Ensure patient safety.

Common Excipients Buffers

- Stabilizers
- Preservatives
- Sugars
- Salts
- Surfactants

Dosage Forms

- Injectable solutions
- Lyophilized (freeze-dried) powders
- Prefilled syringes
- Infusion solutions

7. Manufacturing of Recombinant DNA Drugs

Manufacturing is the large-scale industrial production of recombinant medicines under **Good Manufacturing Practice (GMP)** standards.

Manufacturing Process

1. Cell line development
2. Cell culture
3. Fermentation
4. Protein expression
5. Cell harvesting
6. Purification
7. Formulation
8. Sterile filling
9. Packaging
10. Quality control

Quality Control Tests Purity

- Potency
- Sterility
- Identity
- Endotoxin testing
- Stability testing

8. Handling of Recombinant DNA Drugs

Handling involves the proper storage, transportation, and preparation of recombinant drugs to maintain their quality and effectiveness.

Guidelines

- ✓ Store at **2–8°C** unless otherwise specified.
- ✓ Protect from direct light.
- ✓ Do not freeze if prohibited.
- ✓ Avoid vigorous shaking (especially protein drugs).
- ✓ Maintain sterile conditions.
- ✓ Follow expiry dates.

Safety Precautions

- ➢ Use aseptic techniques.
- ➢ Wear protective equipment (gloves, lab coat).
- ➢ Dispose of biological waste properly.
- ➢ Follow biosafety guidelines.

9. Dispensing of Recombinant DNA Drugs

Dispensing is the process of preparing and supplying recombinant DNA drugs to patients according to a valid prescription while ensuring correct dosage, storage, and patient counseling.

Responsibilities of the Pharmacist

- ✓ Verify the prescription.
- ✓ Check patient identity.
- ✓ Confirm the correct dose.
- ✓ Maintain the cold chain if required.
- ✓ Educate the patient on proper administration and storage.

- ✓ Monitor for adverse reactions.

Patient Counseling Points

- ➢ Follow the prescribed dosage exactly.
- ➢ Store the medicine as instructed.
- ➢ Do not use expired medication.
- ➢ Report allergic reactions or unusual side effects.
- ➢ Do not share biologic medicines with others.

Advantages of rDNA Drugs High purity and quality.

- ❖ Highly specific therapeutic action.
- ❖ Lower risk of contamination compared with animal-derived products.
- ❖ Can replace missing human proteins.
- ❖ Useful for treating many chronic and genetic diseases.

Limitations of rDNA Drugs Expensive to produce.

- ➢ Often require refrigeration (cold chain).
- ➢ Usually administered by injection.
- ➢ May trigger immune responses in some patients.
- ➢ Complex manufacturing and strict quality control requirements.

Summary

- **Cell-based therapy:** Uses living cells to treat or repair diseases.
- **Stem cell therapy:** Uses stem cells to regenerate damaged tissues.
- **Transplant rejection:** Immune response against transplanted cells or tissues.
- **Cell encapsulation technology:** Protects transplanted cells within a semipermeable membrane to reduce immune rejection.
- **Production of rDNA drugs:** Uses genetic engineering to produce therapeutic proteins.
- **Formulation:** Converts purified proteins into stable dosage forms.
- **Manufacturing:** Large-scale GMP production of recombinant medicines.
- **Handling:** Proper storage and preparation to maintain drug quality.
- **Dispensing:** Safe distribution of recombinant medicines with appropriate patient counseling.

END OF GROUP ASSIGNMENTS !!!

10Q 4Watchings!!